AI judges measured in human-rater units

Research discussion | 21 September 2026 | Goal: an Arena-style judge leaderboard

How many human judgments does a specified human benchmark need to match one AI forecast's expected score? The rating is specific to the task, human population, scoring rule, and combiner.

How the measurement works

Build a score curve using k human labels to predict a disjoint human. Place the AI score on that curve and report the first interpolated crossing.

What the number does not mean

0.8 humans is not 80% accuracy or 80% intelligence. Zero means the fitted prior already reaches the AI score. Above-range values remain bounds.

What we have established

MT-Bench	One-judge demonstration: 0.94 humans; interval spans one.
Arena	Calibration improves scores; one vote per battle prevents measuring positive equivalence.
DICES	Repeated humans + 16 configurations; calibration produces three positive point estimates.
AlpacaEval	648 pairs with four humans each; matching AI inference is still pending.
Software	Substantial measured CPU speedups; restricted paths tested; broader audit remains open.

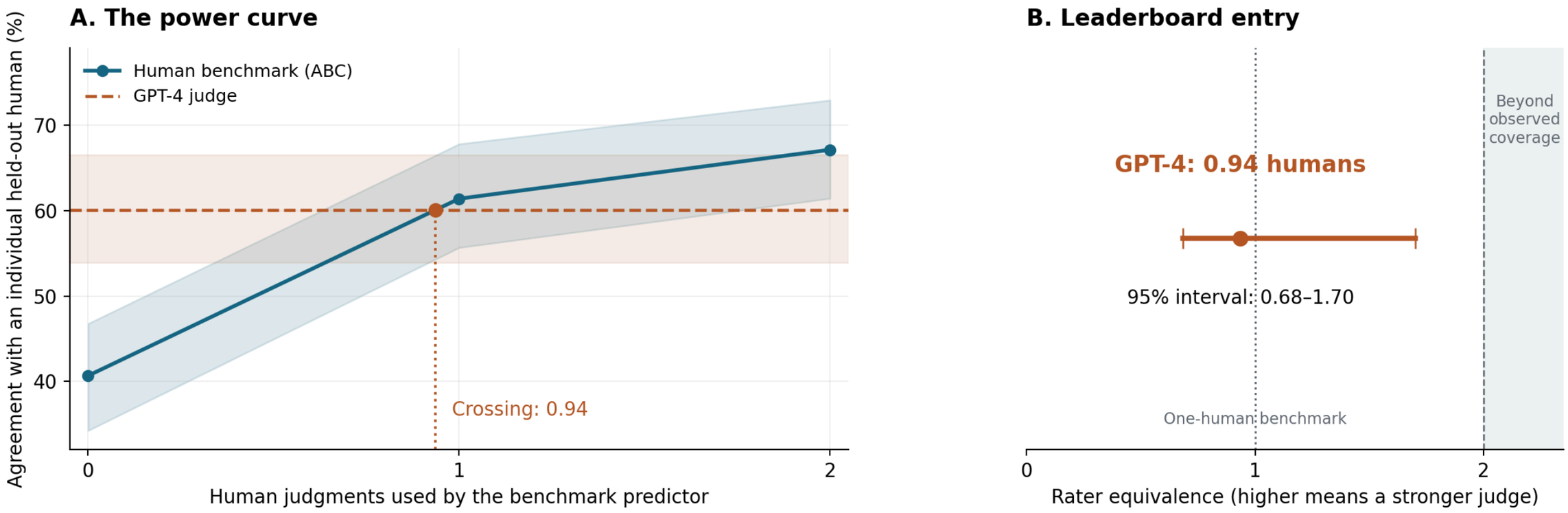
Short presentation: pages 2, 5, 6, 7, then page 9. Pages 3, 4 and 8 provide supporting investigations.

MT-Bench: the intended curve-to-leaderboard view

227 exact items | 3 experts per item | 57 question clusters | Released GPT-4 hard judgments

Human-equivalent judging: show the crossing and its uncertainty

Observed MT-Bench results · GPT-4 as a preference judge · Agreement score



GPT-4's point estimate matches a 0.94-human benchmark. The interval spans one human.

227 exact items; 57 question clusters; 3 selected external experts per item. 500 paired cluster bootstraps.

Shading in A shows pointwise score intervals; B uses paired equivalence replicates. 1 replicate(s) exceed the two-human range and are retained.

Exploratory: independence of released human votes from GPT-4 feedback is unverified. Intervals condition on selected votes and the fitted combiner.
Only one AI judge has this repeated-human comparison. The 16-judge Arena release has one human vote per battle; positive human-panel equivalence is not identified.

TAKEAWAY Agreement equivalence is 0.94 humans (conditional 95% interval 0.68-1.70).

LIMIT Exploratory: human-vote independence from GPT-4 feedback is unresolved. The interval spans one human; one censored bootstrap is retained.

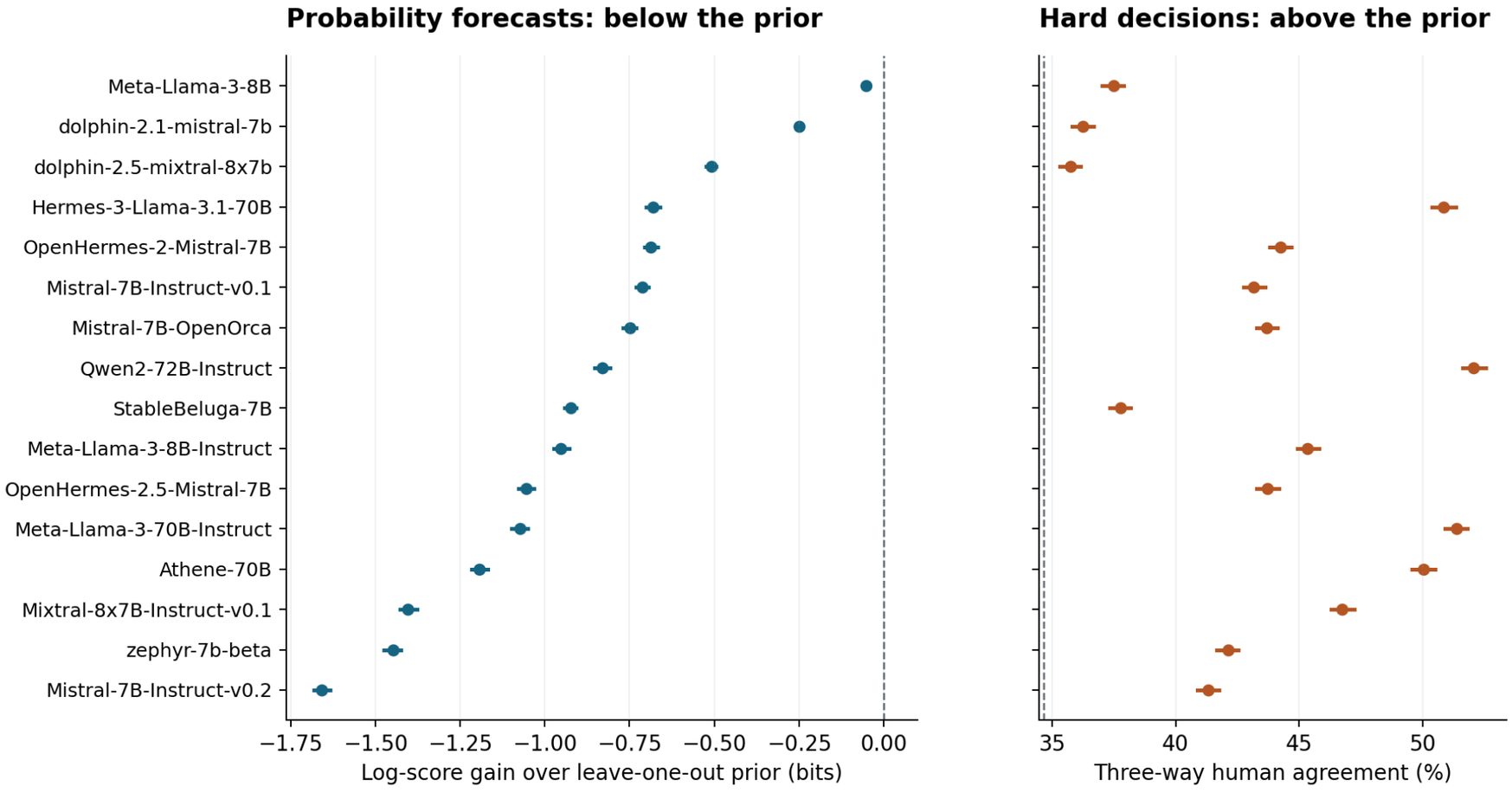
Source: benchmarks/results/paper-protocol/mt_check.json; paper-audit/figure_provenance.json

Arena: useful scores, insufficient human-panel depth

49,938 exact single-turn battles | 16 AI judges | One observed human vote per battle

Arena judges: the scoring rule changes the conclusion

Under the paper’s definition, 16 of 16 released forecasts have empirical log-score equivalence 0.



49,938 exact battles · 95% intervals from 500 paired prompt-cluster bootstraps · Dashed lines: zero-human prior

Left: uncalibrated softmax probabilities, clipped at .02 and renormalized as in the existing experiment. Paired intervals compare each forecast with the same prior.
Right: released strict-winner/tie decisions. Positive human-panel equivalence cannot be measured with one human vote per battle.

This ranks preference judges, not the answer-generating chatbots. Zero log-score equivalence does not mean zero agreement or no useful information.

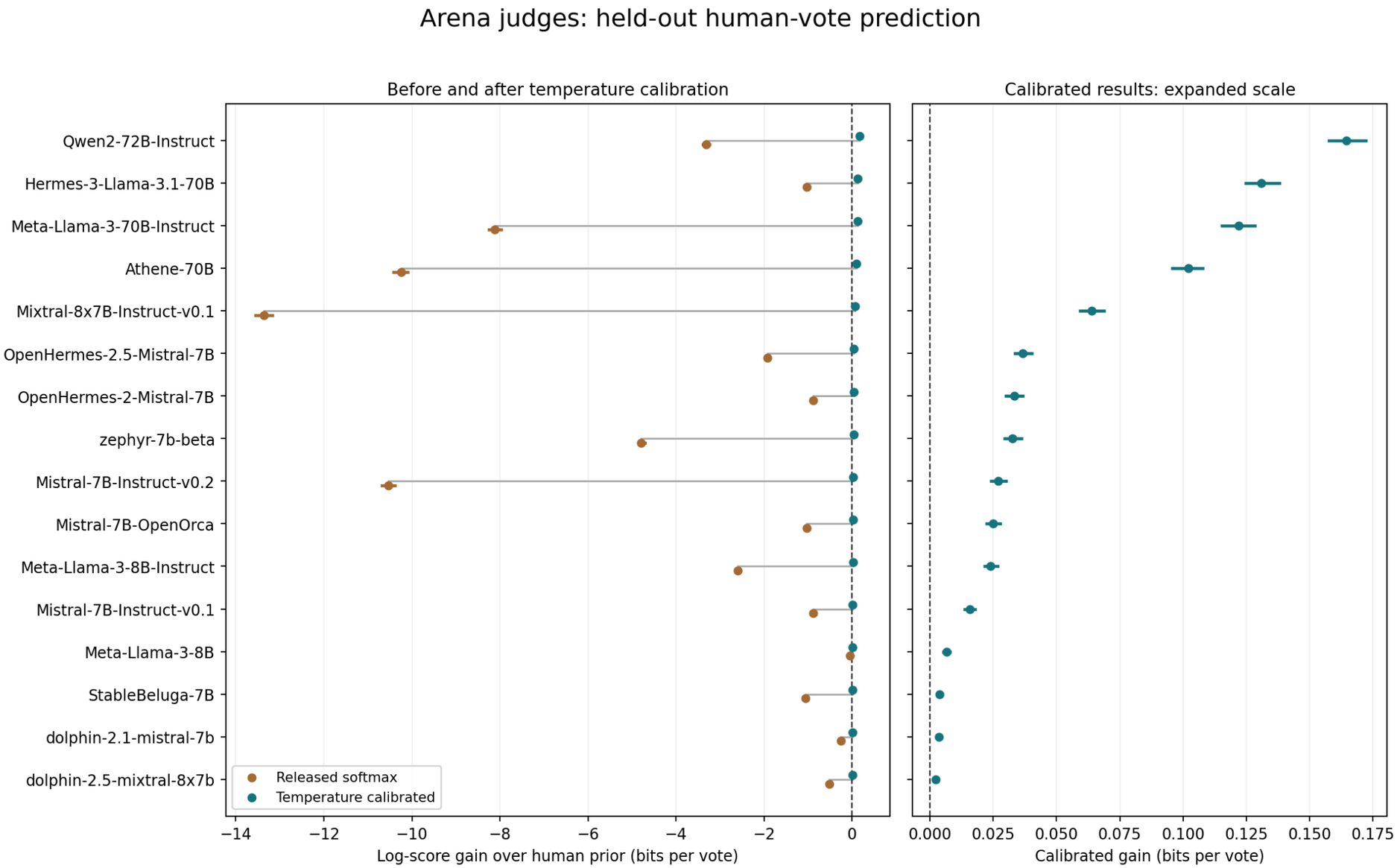
TAKEAWAY Uncalibrated log-score equivalence is zero for all judges; hard Agreement exceeds the prior.

LIMIT These are different scoring questions. Positive equivalence cannot be measured without repeated independent votes on each exact response pair.

Source: benchmarks/results/paper-audit/arena_zero_boundary.json; docs/arena_validation_results.md

Arena calibration: gains in bits are not human counts

14,919 calibration battles | 35,019 evaluation battles | Exact prompt groups kept disjoint



95% paired prompt-bootstrap intervals; fitted temperatures and prior held fixed.
One human vote per battle: this is a calibration diagnostic, not a leaderboard in human-rater units.

TAKEAWAY All 16 calibrated forecasts beat the prior. Qwen2-72B-Instruct gains 0.16455 bits.

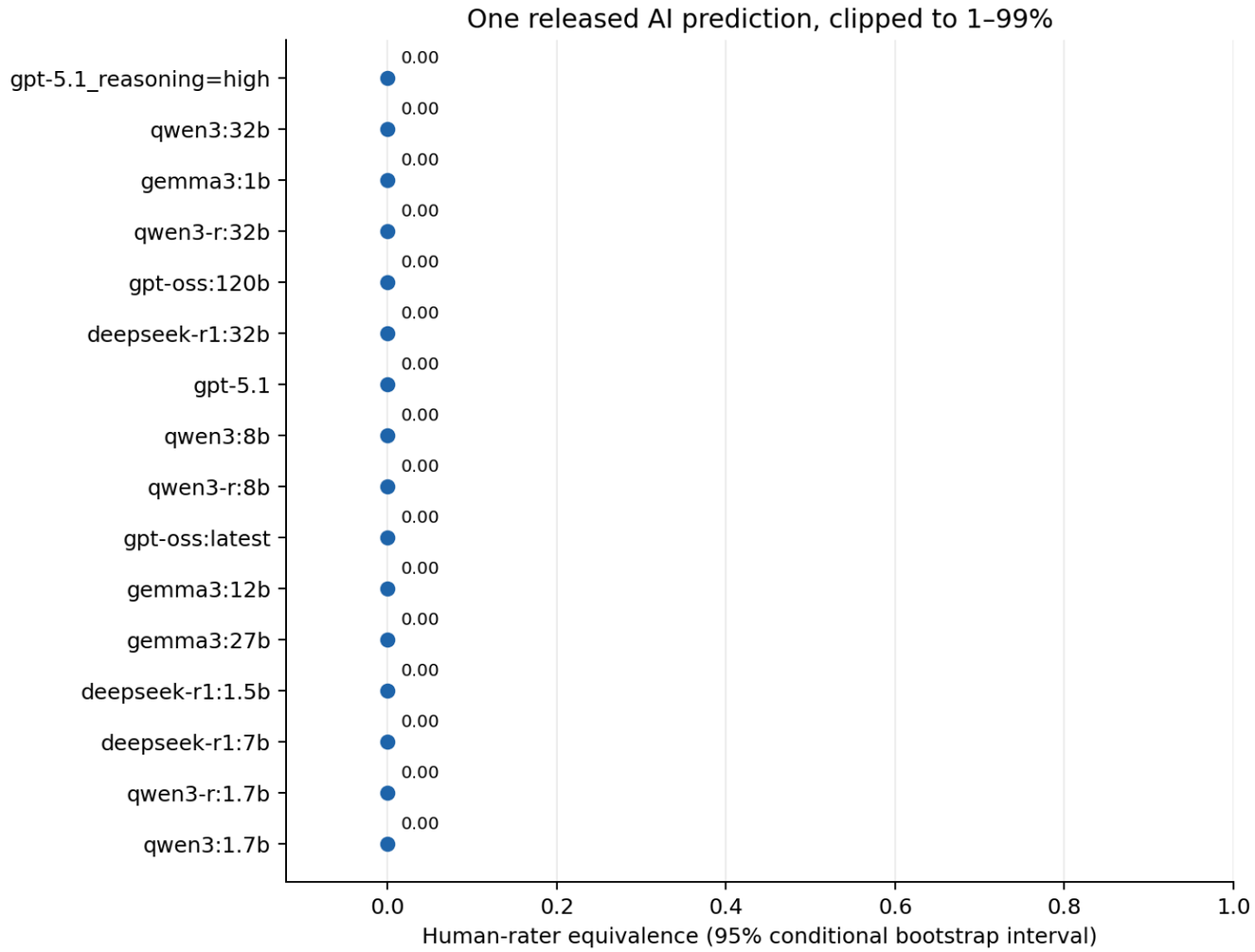
LIMIT Positive human equivalence remains unmeasured. Retrospective temperature calibration; bootstrap holds the fitted temperatures and prior fixed.

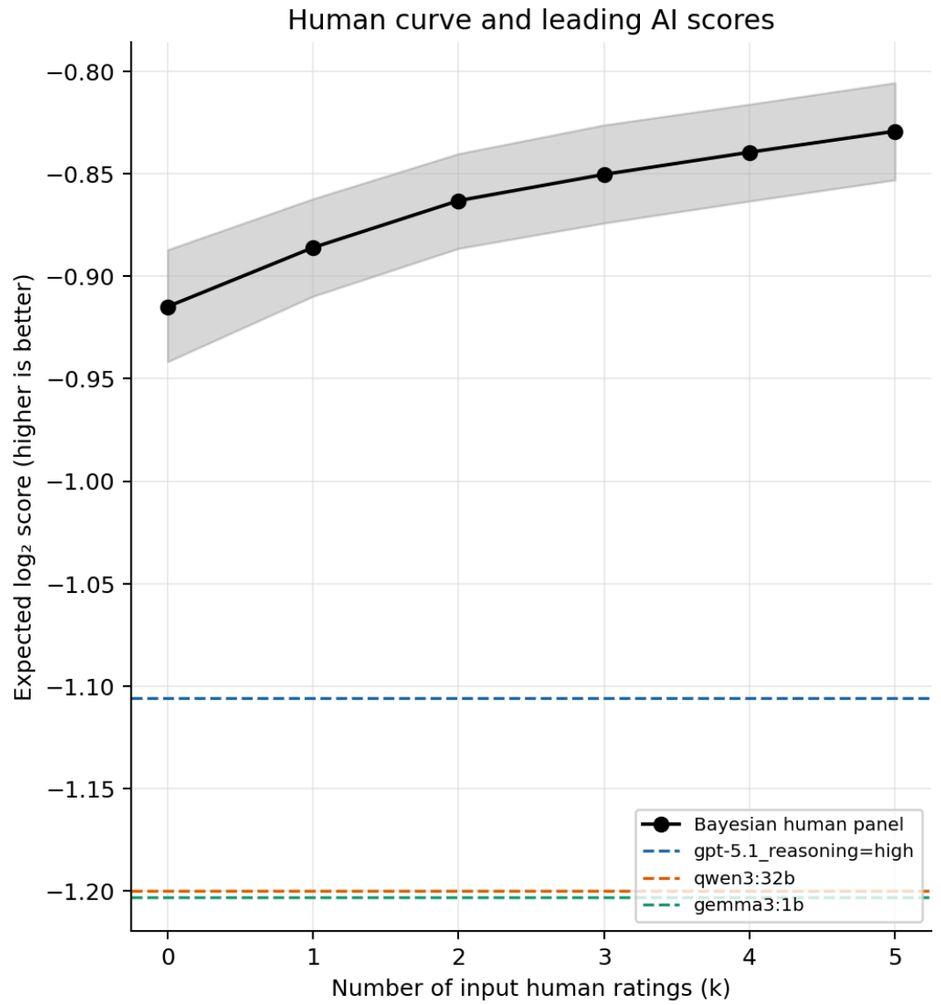
Source: benchmarks/results/arena-calibration/results.json

DICES: a repeated-human, multi-judge experiment

350 conversations | 64 quality-filtered, college-educated humans | 16 released AI configurations

DICES safety-flag prediction • 350 conversations • college-educated human reference group





Equivalence is task-, population- and scoring-specific. Zero means no advantage over the k=0 prior.
Intervals condition on the human pool, fitted combiner, sampled panels and generation 0; ranks are descriptive.

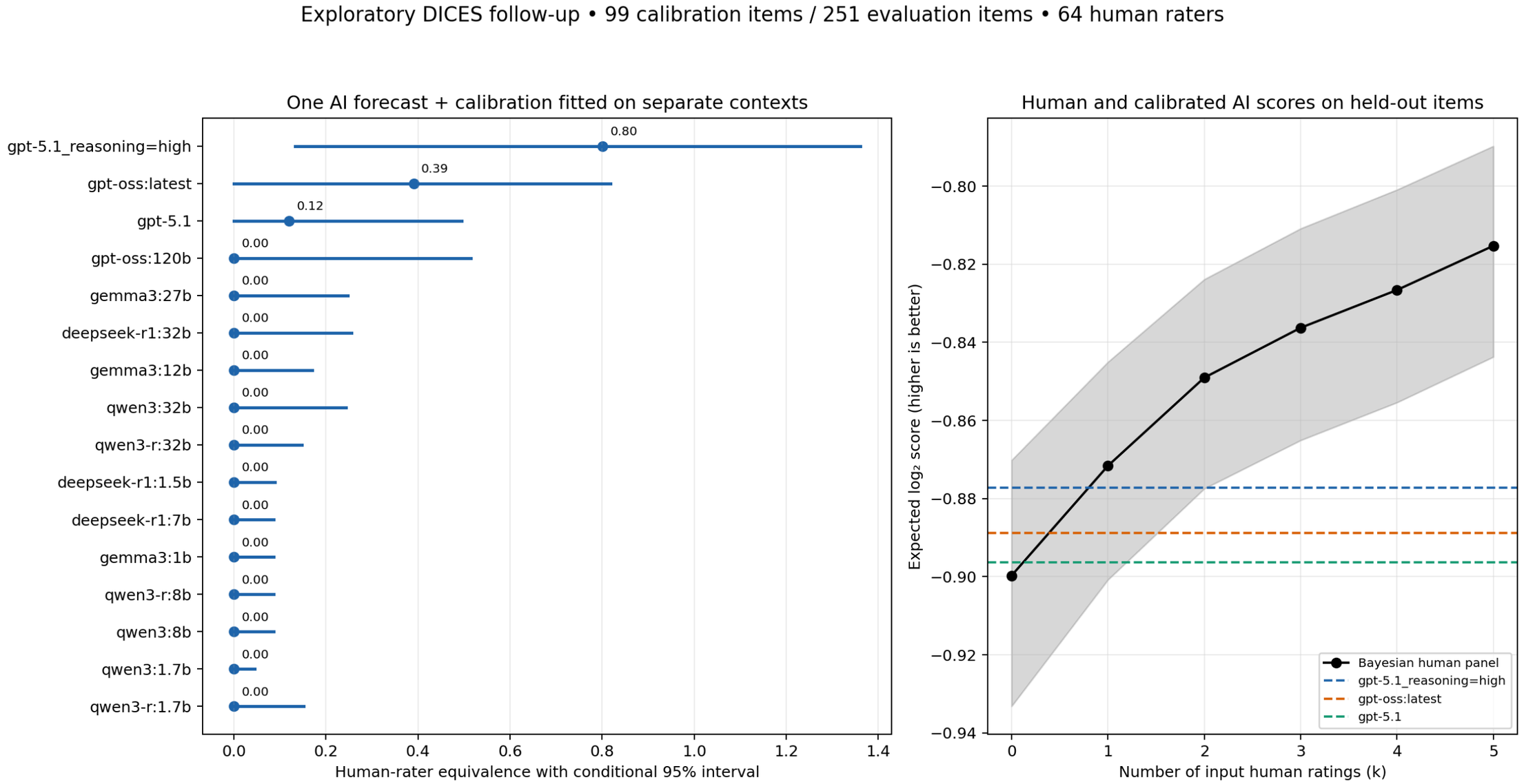
TAKEAWAY All 16 are tied at zero log-score equivalence for the specified 1% probability-floor forecasts.

LIMIT Zero is relative to the prior, not proof of no information. This is safety-flag prediction for this demographic, not general answer-quality Elo.

Source: benchmarks/results/dices-validation/leaderboard.csv; docs/dices_validation_results.md

DICES calibration: positive but exploratory estimates

99 calibration items / 251 evaluation items | Disjoint context groups | One forecast + fitted calibration



This evaluates the calibrated judge system, not the original raw forecasts. Calibration and evaluation contexts are disjoint.
Intervals condition on fitted calibration, human pool, fitted combiner, sampled panels and generation 0; ordering is descriptive.

TAKEAWAY Leading estimates: 0.801 humans (0.134-1.361), 0.391 (0-0.819), and 0.119 (0-0.495).

LIMIT Calibration was motivated by inspecting full-data results. This is not confirmatory; it does not establish superiority over one human or a definitive ranking.

Source: benchmarks/results/dices-calibration/leaderboard.csv; dices-calibration/manifest.json

AlpacaEval: ready to evaluate general-preference judges

648 exact response pairs | Four selected distinct humans per pair | 605 prompt clusters



Human calibration curves only — matching AI judgments have not been generated.
95% paired prompt-cluster intervals, conditional on the fixed human panel and fitted combiner.

TAKEAWAY The human baseline and 648 blinded Astra requests are ready. No AI-equivalence result exists yet.

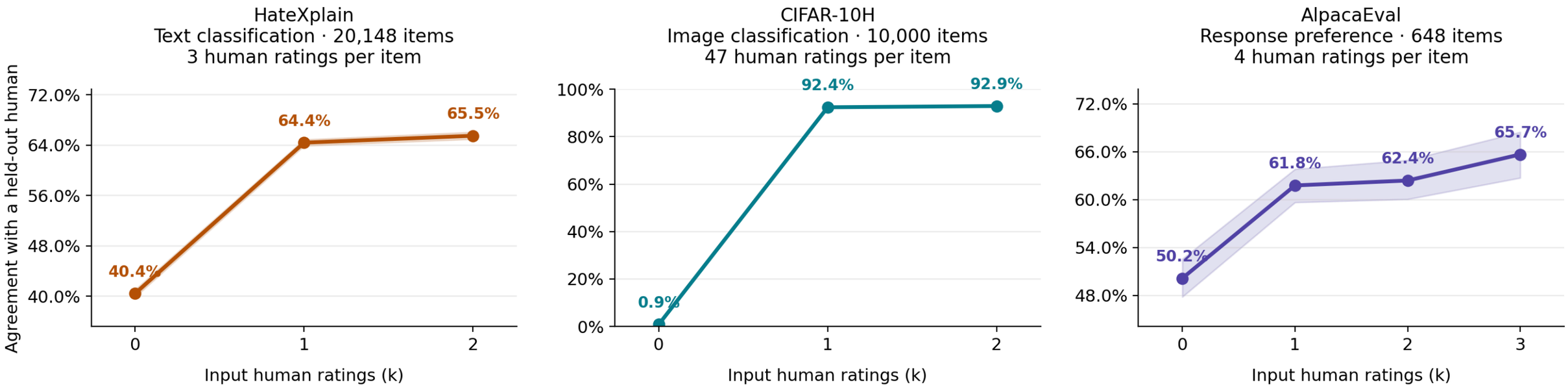
LIMIT Agreement reaches 65.66% at k=3. Matching AI forecasts must be generated; the released GPT-4 file overlaps only four exact pairs.

Source: benchmarks/results/alpaca-human-baseline/manifest.json; docs/alpaca_judge_validation.md

Candidate datasets: human-only feasibility checks

HateXplain: 20,148 posts | CIFAR-10H: 10,000 images | AlpacaEval: 648 response pairs

Human-panel benchmarks for three rater-equivalence tasks



Shading: 95% conditional bootstrap intervals. Each panel uses its own vertical scale.
Astra is not plotted: matching predictions are not available. These curves are human baselines, not AI equivalence estimates.

TAKEAWAY Repeated-human data support curves on several tasks; these figures contain no new AI scores.

LIMIT CIFAR-10H k=0 is 0.87%: a leave-one-out class-prior artifact, not chance accuracy. Use an independent-training sensitivity check. Scales differ by panel.

Source: benchmarks/results/dataset-human-curves/curve_summary.csv; dataset-human-curves/README.md

What is validated, what remains, and the next decision

Do not turn a passing regression suite into a claim of complete mathematical or statistical certification.

Implementation and paper audit

Repaired missing/index alignment, factorial overflow, fixed-class AUC, named-label DMI, training-item identity, explicit clipping and equivalence boundaries. The leave-one-out expectation correction was independently derived.

What remains unresolved

Nonlinear expected-DMI interpretation; other scorer conventions; numeric/padded inputs; reliability and ratio indices; legacy interval plotting; some execution/persistence paths. Research plots use separate reporting helpers.

Performance and verification

Measured ABC/log-score pipeline speedups: 10.15x, 17.56x, 51.27x for 50/1,000/2,000 items. Three repetitions against the corrected reference; fixed small panel range. 109 core + 25 research tests pass. No GPU result.

Uncertainty and interpretation

Reported intervals are conditional on fitted models and selected humans; they omit several sources of refitting/population uncertainty. MT-Bench 80% agreement claims depend on tie policy and cohort. ChaosNLI had no completed run.

Recommended next step: a prospectively specified AlpacaEval judge comparison

1. Agree on the human population, primary score, judge list, rubric, and calibration policy.
2. Freeze common pairs and policies. If calibrating, split prompt groups before examining evaluation outcomes.
3. Generate one forecast per item; evaluate every judge on the same panels and paired bootstrap samples.
4. Show supported crossings and paired comparisons; report >3 rather than extrapolating.

Professor decision: prioritize the general-preference leaderboard, or confirm the calibrated safety result on a fresh cohort?
Evidence: docs/research_professor_briefing.md; benchmarks/REPORT.md; docs/arena_leaderboard_progress.md